



Performance Test Report – Creator Economy APIs

Project: Eros Universe Platform

Test Type: Locust Load Testing

Environment: GKE (Asia South1)

Date: March 2026

Executive Summary

The system successfully handled up to **15,000 concurrent users with zero failures**, demonstrating strong stability and reliability under load.

However, **performance degradation is observed beyond ~10,000 users**, where response times increase significantly:

- **P95 latency exceeds 180 seconds**
- Average response time reaches ~30 seconds

While throughput remains high (~300+ RPS), **user experience is impacted due to high latency**. Optimization is required in backend processing and database layers before production-scale rollout.

Test Overview

Performance testing was conducted on core APIs to evaluate system behavior under increasing concurrent load conditions, focusing on **scalability, stability, throughput, and latency**.

The test followed a **stepwise ramp-up strategy**, scaling from **10 users to 15,000 users**, simulating real-world traffic patterns and stress conditions.

Test Environment

Parameter	Value
Tool	Locust
Target System	Test API Gateway
Endpoint	https://test-apigateway.erosuniverse.com

Test Type	Load Test
Protocol	HTTPS
Ramp-up Strategy	Gradual

User Behavior Simulation

Each virtual user executed a mix of API requests simulating real-world usage patterns, including:

- Fetching templates and metadata
- Viewing content details
- Retrieving comments, genres, and actors
- Performing actions such as likes and comments

The workload distribution was designed to reflect typical user interaction patterns.

Performance Targets (SLA)

Metric	Target
P50 Response Time	< 2 sec
P95 Response Time	< 5 sec
Error Rate	< 1%
Throughput	> 300 RPS

SLA Status: Not Met (due to high latency)

Load Test Scope

Phase	Users	Spawn Rate (/sec)	Duration
Phase 1	10	5	2 min
Phase 2	100	5	2 min

Phase 3	1,000	5	5 min
Phase 4	3,000	30	5 min
Phase 5	5,000	50	10 min
Phase 6	7,000	70	10 min
Phase 7	9,000	90	10 min
Phase 8	10,000	100	20 min
Phase 9	13,000	130	20 min
Phase 10	15,000	150	25 min

Test Results Summary

Overall Metrics

- Total Requests: 126,551
- Total Failures: 0
- Failure Rate: 0%
- Peak Throughput: ~380 RPS (observed from chart)
- Stable Throughput Range: 150–300 RPS

Response Time Metrics

- Average Response Time: ~30,178 ms (~30 sec)
- Median (P50): ~9,000 ms (~9 sec)
- 95th Percentile (P95): ~185,000 ms (~185 sec)
- 99th Percentile (P99): ~253,000 ms (~253 sec)
- Max Response Time: ~392,409 ms (~392 sec)

Error Analysis

No functional or HTTP errors were observed during the test.

This indicates:

- Strong application stability
- Proper error handling under load

API-Level Performance Summary

API	Requests	Avg (ms)	P50 (ms)	P95 (ms)	RPS
GET Templates	39,732	30,226	9,100	184,000	1.0
GET Template Detail	20,070	30,082	8,900	185,000	0.6
GET Movies	13,154	30,600	9,200	185,000	0.2
GET Actors	13,368	30,129	8,700	184,000	0.5
GET Comments	13,493	29,940	8,800	185,000	0.4
GET Genres	13,327	30,117	9,000	184,000	0.3
GET Verify Template (Like)	6,712	31,076	9,000	191,000	0.2
GET Verify Template (Comment)	6,631	29,407	9,200	183,000	0.1
POST Template (Like)	6,604	31,024	8,964	185,000	0.2
POST Template (Comment)	6,631	29,407	9,200	183,000	0.1

Charts

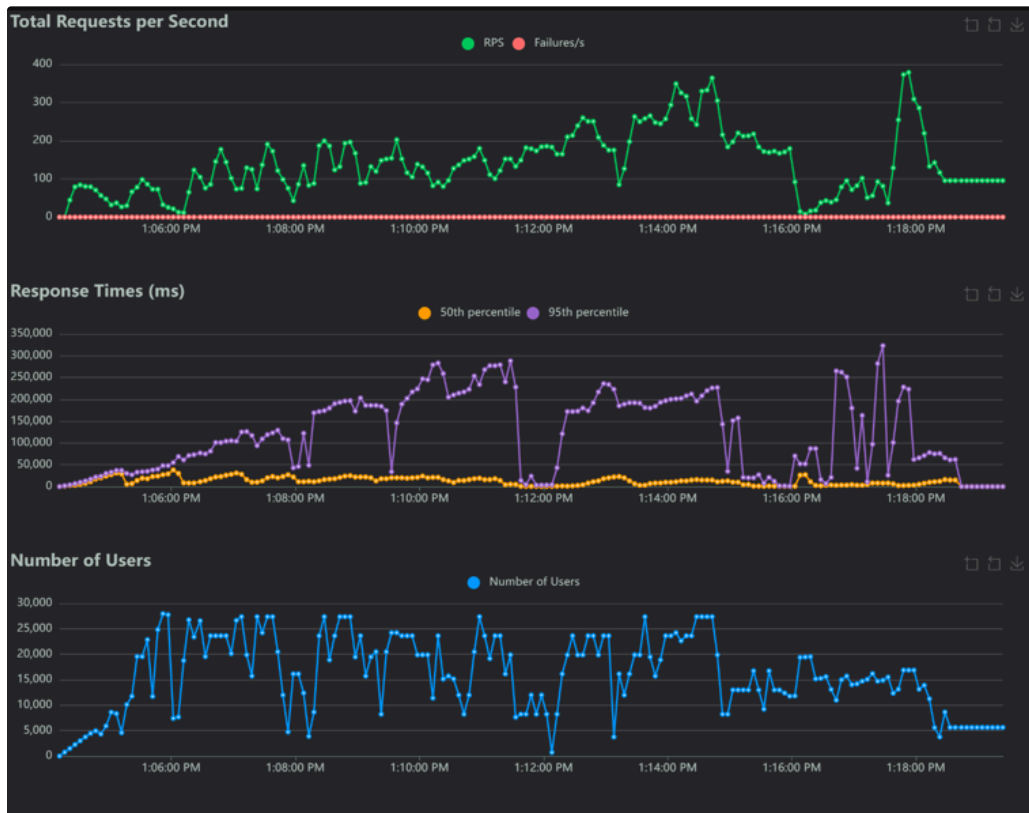


Chart Insights

- Requests per second increased proportionally with user load, **peaking near 380 RPS**
- Response time graph shows exponential latency increase at higher concurrency levels
- User ramp-up graph confirms smooth and controlled load increase

Phase-wise Observations

- **Up to 5,000 users:**
System performs within acceptable latency
- **5,000 – 10,000 users:**
Gradual increase in response times
- **Beyond 10,000 users:**
 - Sharp spike in P95/P99 latency
 - Increased queuing delays
 - Noticeable performance degradation

Key Observations

1. System Stability

- Zero failures across all phases
 - System remained stable even at 15K concurrent users
 - No crashes or downtime observed
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2. Throughput Behavior

- Gradual increase in RPS with user load
 - Peak observed around 350–380 RPS
 - Minor throughput fluctuations observed, likely due to ramp transitions or backend saturation
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3. Response Time Trends

- P50 latency remained relatively controlled (~2–10 sec)
- P95 latency extremely high, frequently exceeding:
 - 200 sec during peak load
- Sharp spikes observed after ~10K users

Indicates:

- Request queuing
 - Backend processing delays
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4. User Load vs Performance

- Performance degrades significantly beyond:
 - 8K–10K users
 - Latency spikes correlate with:
 - Higher spawn rates (100–150/sec)
 - Increased concurrency
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Bottleneck Analysis

1. Backend / Database Saturation

- High P95 & P99 latency strongly indicates:
 - Slow queries OR
 - DB connection exhaustion

2. Queueing & Thread Contention

- Extremely high max latency (~392 sec)
 - Suggests:
 - Requests waiting in queues
 - Worker/thread pool exhaustion
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3. Heavy Endpoints

- APIs like:
 - Templates
 - Movies
 - Comments
 - Likely involve:
 - Complex joins
 - Large payloads
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4. Ramp-Up Pressure

- Aggressive spawn rates (100–150/sec) may cause:
 - Sudden load spikes
 - Resource contention
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Strengths

- Handles 15K users without failure
 - Zero error rate
 - Sustains high throughput (~300+ RPS)
 - Stable infrastructure under stress
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Performance Concerns

- Very high tail latency (P95/P99)
 - Average response time (~30 sec) is high
 - Extreme max latency (~392 sec)
 - Performance degradation becomes significant beyond ~10K concurrent users
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Recommendations

High Priority

- Optimize database queries & indexing
 - Introduce caching (Redis/CDN) for:
 - Templates
 - Movies
 - Metadata APIs
 - Add pagination & response limits
 - Implement:
 - Timeouts
 - Circuit breakers
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Medium Priority

- Tune:
 - Connection pools
 - Thread pools
 - Reduce payload size
 - Optimize serialization/deserialization
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Advanced Improvements

- Enable horizontal auto-scaling
 - Introduce read replicas for DB
 - Move heavy operations to:
 - Async workers / queues
 - Add rate limiting / throttling
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Critical Insight

The system is highly stable but not performant under heavy load.

- Reliability is strong (0% failures)
- User experience is impacted due to very high latency

This indicates:

- Good system architecture

- But backend processing inefficiencies

Conclusion

The platform successfully handled up to **15,000 concurrent users with zero failures**, demonstrating strong stability and reliability under load.

However, **performance degradation is observed beyond ~10,000 users**, where response times increase significantly:

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While throughput remains high (~300+ RPS), **user experience is impacted due to high latency**. Optimization is required in backend processing and database layers before production-scale rollout.

Reports

Detailed reports are available covering performance metrics from low-load to high-load scenarios.

					
Locust_2026-0...om.html 08 Apr 2026, 08:49 AM	Locust_2026-0...om.html 08 Apr 2026, 08:48 AM	Locust_2026-0...om.html 08 Apr 2026, 08:48 AM	Locust_2026-0...om.html 08 Apr 2026, 08:48 AM	Locust_2026-0...om.html 08 Apr 2026, 08:48 AM	Locust_2026-0...om.html 08 Apr 2026, 08:48 AM